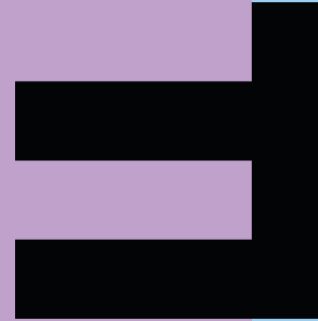


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of Applied Sciences



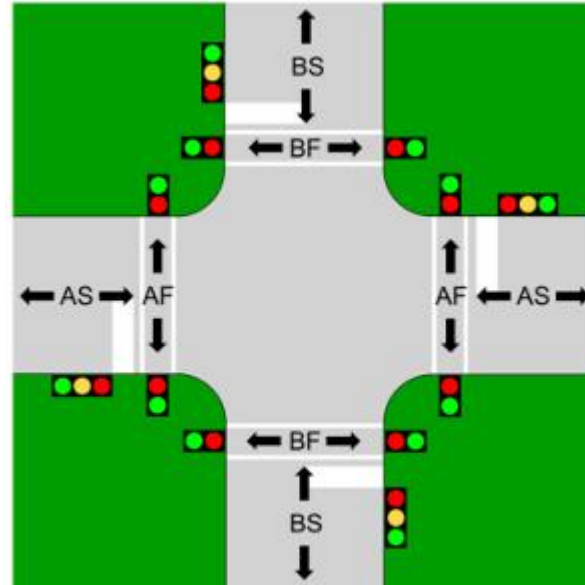
State Machine Example

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Traffic Light Control I

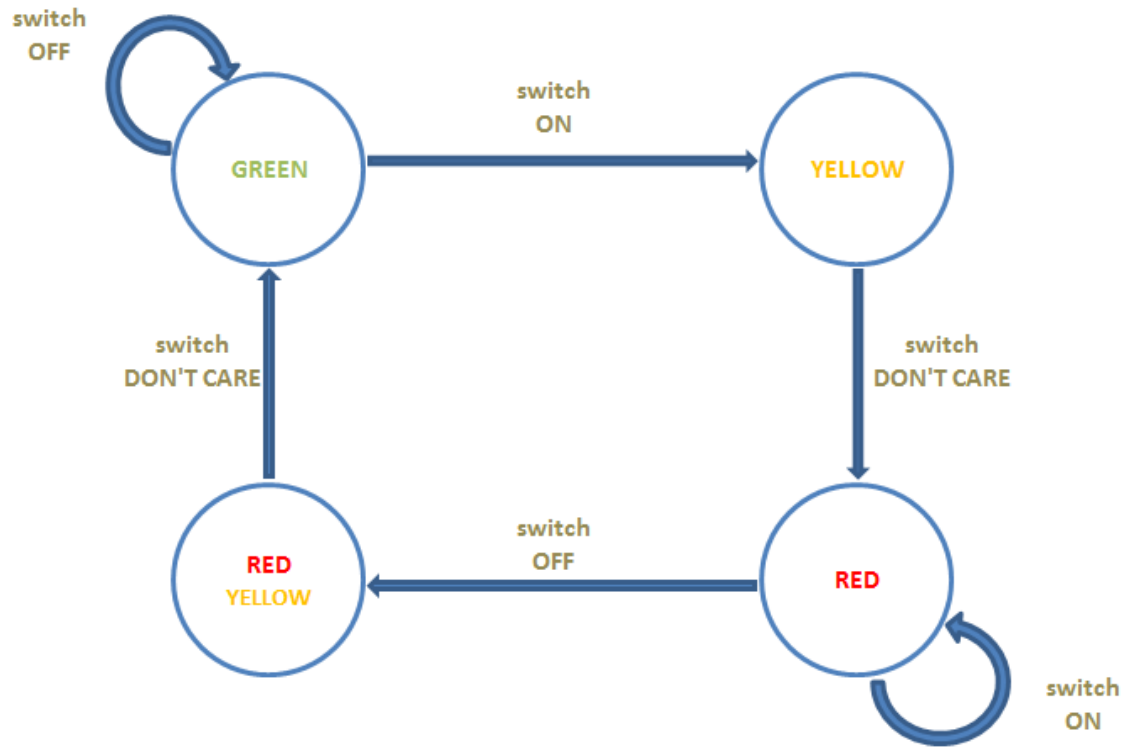


Assumptions

- only one pedestrian traffic light is in the focus
- Switch will start switching process – starting at “green” – ending at “red”
- Switch back will start process again – ending at “green”



Traffic Light Control II

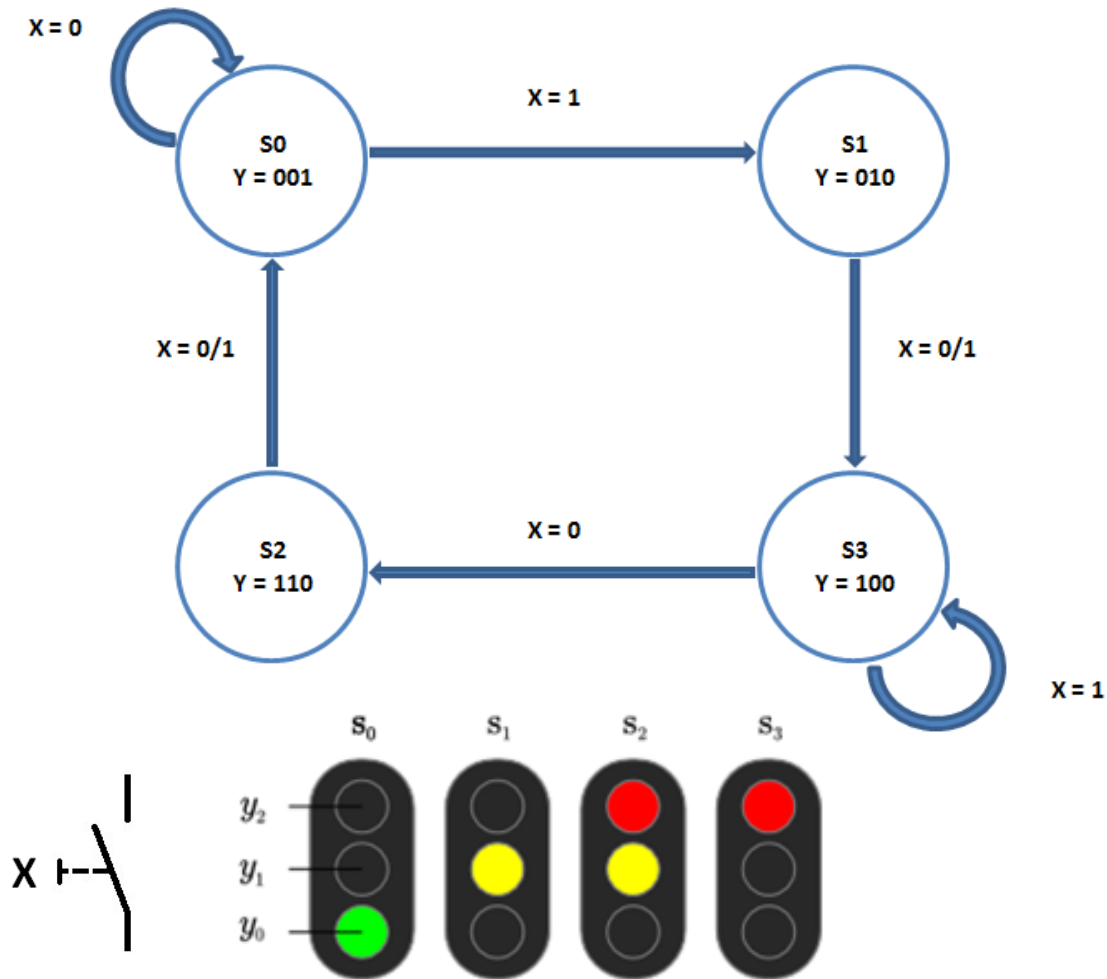


State Machine – Moore type

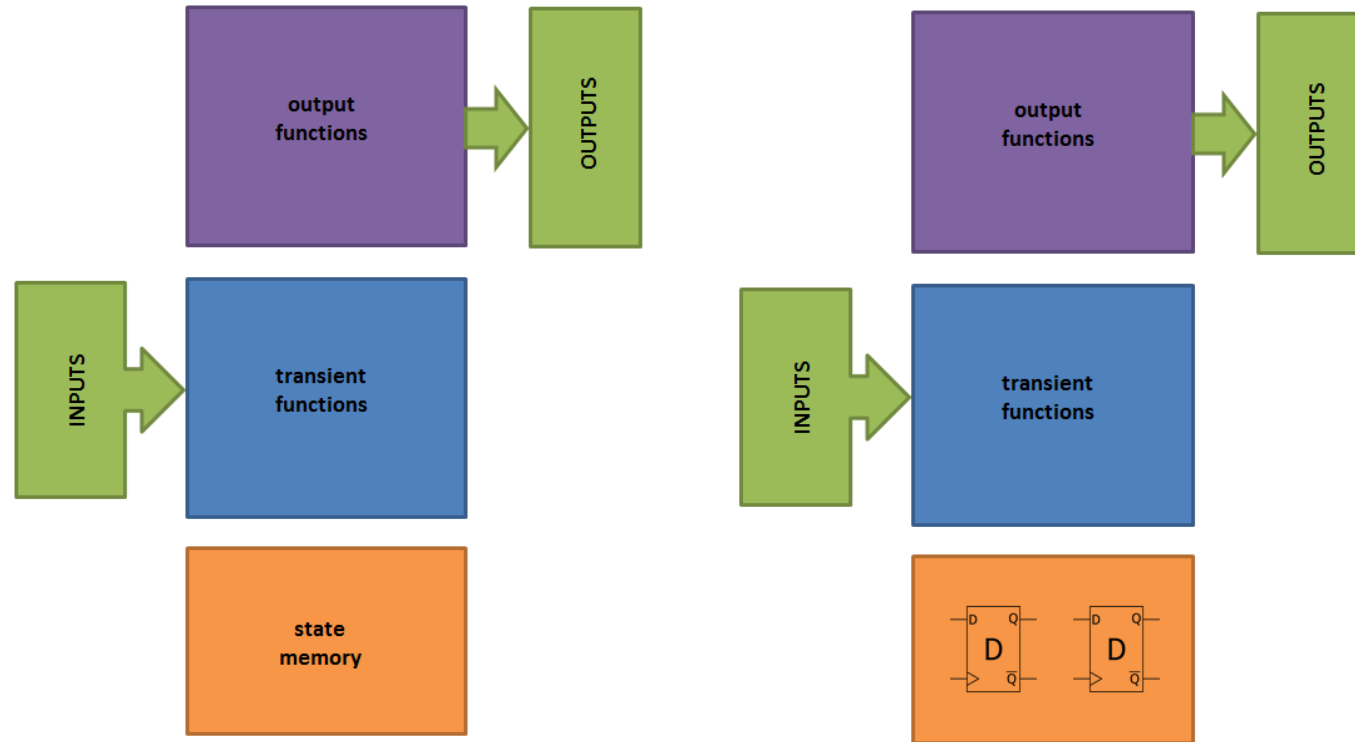
- Outputs only depend on the actual state



Traffic Light Control III



Traffic Light Control IV



Moore machine

- Each state is represented by one D-flip-flop
- Output and transient functions can be analyzed independently



Traffic Light Control V

- Transient functions

state	q_1	q_0	x	d_1	d_0
s 0	0	0	0		
s 0	0	0	1		
s 1	0	1	0		
s 1	0	1	1		
s 2	1	0	0		
s 2	1	0	1		
s 3	1	1	0		
s 3	1	1	1		



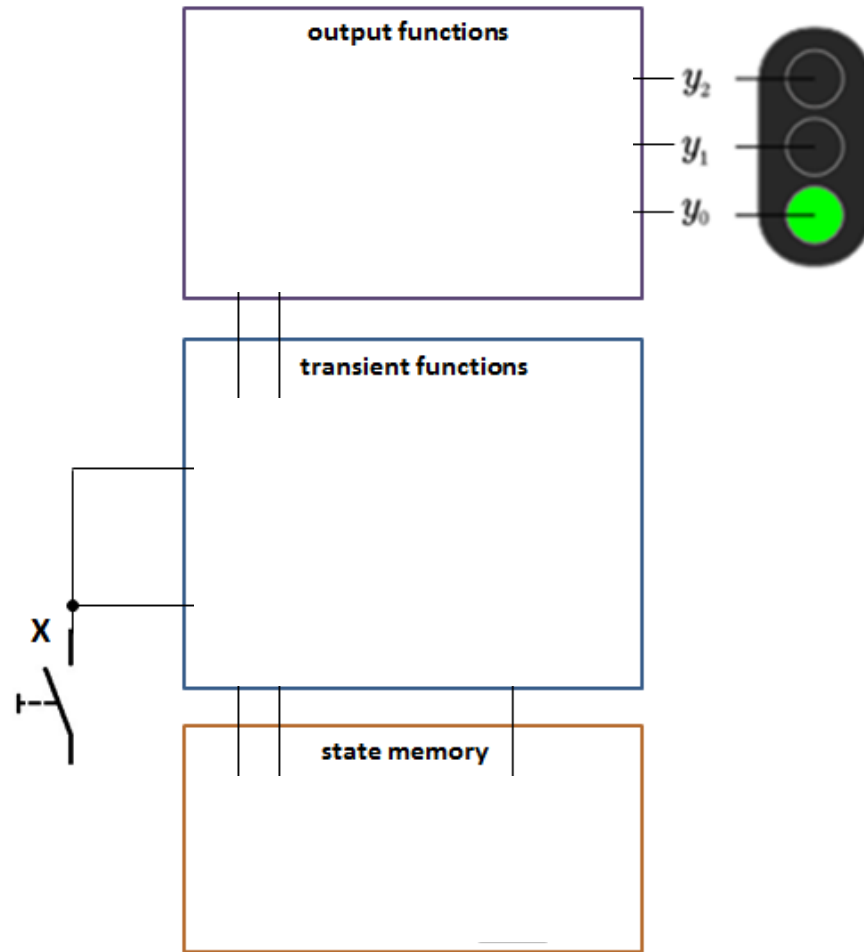
Traffic Light Control VI

- Output functions

state	q_1	q_0	y_2	y_1	y_0
s_0	0	0			
s_1	0	1			
s_2	1	0			
s_3	1	1			



Traffic Light Control VII



State Machine Example

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